HGVRec: Action-Segment Hypergraph Aggregation with Role-aware Entity–Action Fusion for Video Recommendation

Kihoon Kim
josepkk.kim@samsung.com
Samsung Electronics
Suwon, Republic of Korea

Junsu Choi
junsu2.choi@samsung.com
Samsung Electronics
Suwon, Republic of Korea

Doyeol Baek
doyeol.baek@samsung.com
Samsung Electronics
Suwon, Republic of Korea

Hyung Rae Kim
hyungrae77.kim@samsung.com
Samsung Electronics
Suwon, Republic of Korea

Jieun Kang
jauthor.kang@samsung.com
Samsung Electronics
Suwon, Republic of Korea

Doyun Choi
doyun.choi@snu.ac.kr
Seoul National University
Seoul, Republic of Korea

Jiwon Seo
jiwon.seo@snu.ac.kr
Seoul National University
Seoul, Republic of Korea

Jaemin Yoo*
jaeminyoo@snu.ac.kr
Seoul National University
Seoul, Republic of Korea

Abstract

Multimodal large language models (MLLMs) provide high-level video semantics that can complement metadata and interaction signals in recommendation. Privacy concerns around server-side analysis of viewing content motivate on-device processing with lightweight MLLMs. Under constrained observation and compute budgets, however, their scene descriptions may be incomplete or inaccurate. The challenge is to organize the extracted information into representations useful for recommendation. We introduce HGVRec, a video representation learning framework that uses explicit structure to improve the recommendation utility of MLLM-generated scene graphs. Action-Segment Hypergraph Aggregation (ASHA) models scenes as nodes and maximal contiguous runs sharing a normalized action key as hyperedges; scenes containing multiple actions can participate in overlapping segments. ASHA aggregates information from scenes to segments and back, and incorporates segment extent into video readout. Role-aware Entity–Action Fusion (REAF) enriches action features with entity features according to extracted actor, target, tool, and place roles, providing the scene representations aggregated by ASHA. These explicit segment memberships and role links make the modeled structure inspectable. The representation modules are trained jointly with a sequential recommender using frozen text embeddings of the extracted scene information. By separating semantic extraction from recommendation learning, HGVRec accommodates both lightweight and more capable MLLMs, while using action segments as explicit aggregation units for next-video recommendation.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Learning latent representations.*

Keywords

Video Recommendation, Hypergraph Learning, Scene Graph, Multimodal Large Language Models

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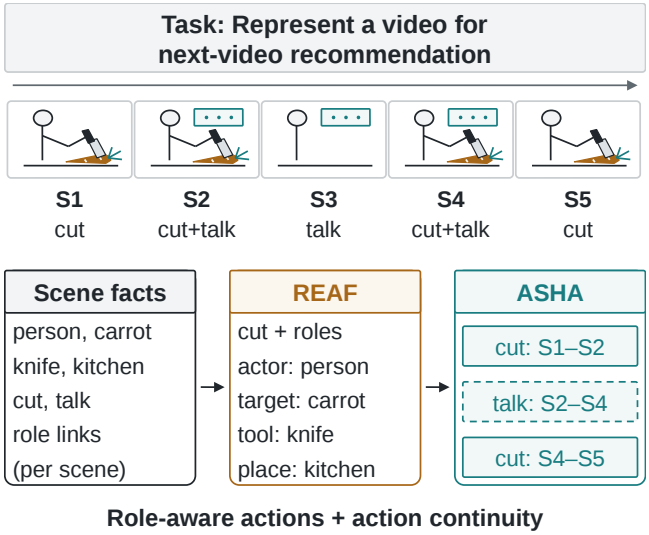


Figure 1: An illustrative example of HGVRec. REAF encodes who acts on what and with which tool, while ASHA groups consecutive scenes sharing an action. Both use the same extracted scene information; the two cut segments remain separate across the interruption.

*Corresponding author.

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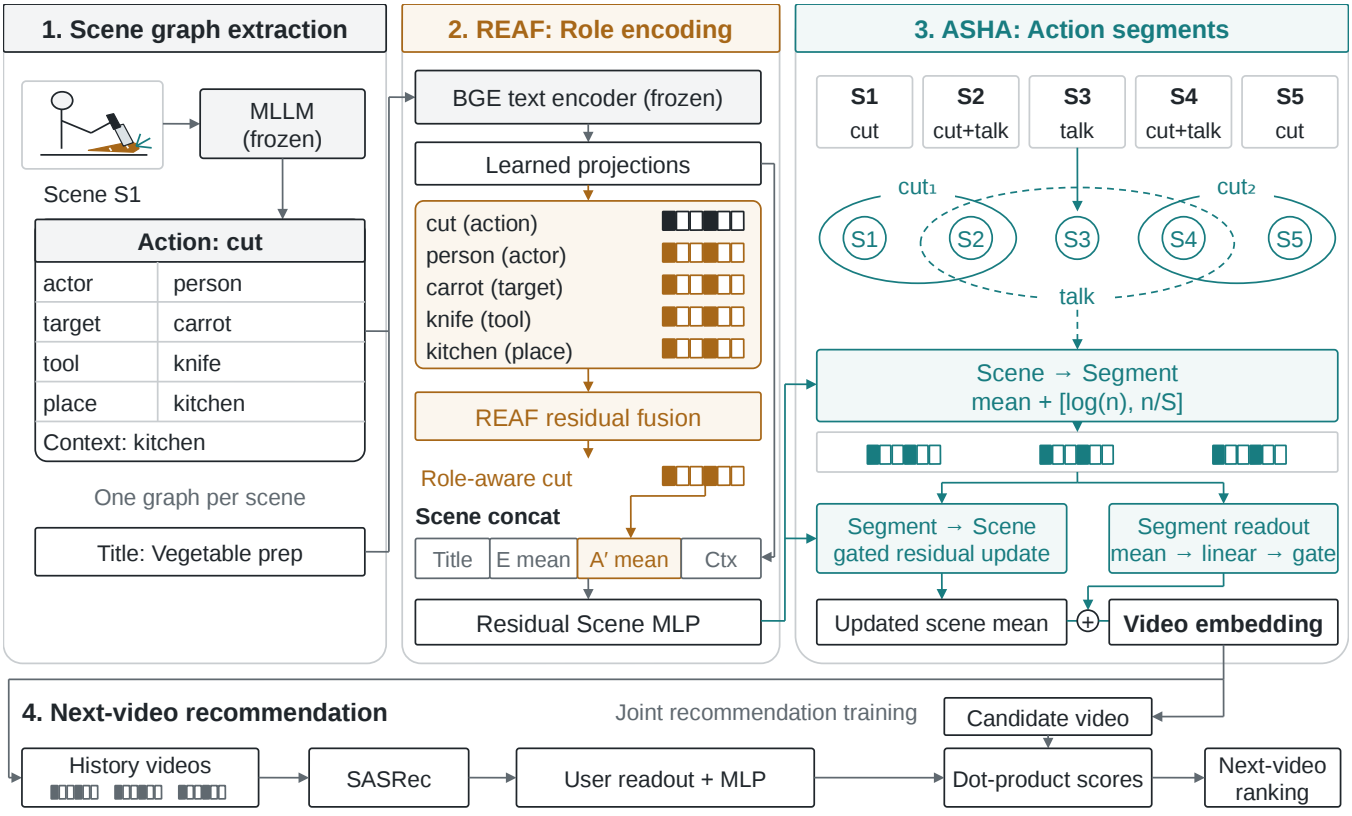


Figure 2: Overview of HGVRec using the same cooking example. (1) A frozen MLLM extracts scene graphs. (2) Frozen text embeddings and learned projections feed REAF, which fuses action features with their role-linked entities. (3) ASHA aggregates contiguous action segments and combines scene and segment readouts. (4) The resulting video embeddings support next-video recommendation. Vector cells are schematic; n and S count segment scenes and valid video scenes, respectively.

1 Introduction

Video recommendation on Web platforms depends on representations that connect visual content with user preferences. Multimodal large language models (MLLMs) provide semantic information beyond item metadata: DWYS [3] uses generated descriptions as recommendation inputs, while IISAN [5] adapts multimodal representations through side networks. Resource constraints can favor compact MLLMs for content extraction. This raises a representation learning question: how can a recommender make better use of the information that a compact MLLM already extracts?

We investigate explicit structure as a way to improve the recommendation utility of compact MLLM outputs. A scene may contain entities and actions, yet their separate aggregation leaves participation roles implicit; likewise, aggregating scene vectors without action-level organization leaves continuity implicit. The opportunity is to expose these relations to the recommender through structured representations of the available observations.

For example, a person may cut vegetables with a knife across several consecutive scenes. The person, vegetables, and knife play different roles in the same action, while the repeated action links the scenes into a coherent segment. This suggests contiguous action segments as aggregation units for video representation. We examine this choice within each video; user interaction histories remain modeled by a sequential recommender.

We introduce HGVRec, a hierarchical framework centered on Action-Segment Hypergraph Aggregation (ASHA). Building on the general aggregation framework of AllSet [2], ASHA treats scenes as nodes and maximal contiguous runs sharing a normalized action key as hyperedges. Scenes with multiple actions can belong to overlapping segments. Messages pass from scenes to segments and back, with segment extent incorporated into the representations and video readout. Role-aware Entity–Action Fusion (REAF) supplies ASHA with scene features by fusing entity features into actions according to extracted actor, target, tool, and place assignments. These explicit memberships and role links make the modeled structure inspectable.

We frame evaluation around structural modeling with a fixed compact MLLM, comparisons with scene sequence and attention encoders, and the use of more capable extractors. As device chips improve or server AI becomes available, stronger MLLMs may become practical; HGVRec's modular extraction stage allows their recommendation benefits to be investigated within the same framework. Our contributions and the empirical questions they motivate are:

- **Structured representations from compact MLLMs.** We propose a framework that explicitly organizes extracted semantics into role-aware scenes and action segments to improve their utility for recommendation.
- **Action segments as aggregation units.** We formulate contiguous action grouping as a video representation unit and investigate its utility relative to chronological scene encoding.
- **Inspectable relational modeling.** ASHA and REAF define explicit scene memberships and role links, supporting structural inspection and comparison with attention-based scene aggregation.
- **Extension across MLLM capabilities.** The framework separates semantic extraction from recommendation learning, enabling evaluation of compact and more capable MLLMs under a common recommendation setup.

2 Related Works

Video recommendation benefits from modality-specific graph representations, as in MMGCN [14], and MLLM-generated descriptions, as in DWYS [3]. DFF [12] further studies representations from frozen MLLMs and their fusion with item IDs. Hypergraphs also support recommendation: DHCN [15] connects items through sessions, and MHCR [8] combines multiple graph views with hypergraph contrastive learning. HGVRec constructs a hypergraph within each video, with scenes as nodes and contiguous action segments as hyperedges.

Hypergraph message passing is well established by HGNN [4] and the multiset aggregation framework of AllSet [2]. ED-HNN [13] introduces receiver-dependent hyperedge messages, and ZEN [1] unifies linearized forms of representative hypergraph networks. ASHA follows the general pattern of aggregation between nodes and hyperedges, using action continuity to define overlapping groups of scenes and incorporating segment extent into aggregation and video readout.

Structured visual representations encode actions through participants and relations: Grounded Situation Recognition [10] models entities in semantic roles, while Action Genome [6] represents actions through spatiotemporal scene graphs. REAF uses extracted actor, target, tool, and place assignments to fuse entity features into action representations for recommendation, supplying role-aware scene features to ASHA.

3 Methods

3.1 Problem Formulation and Overview

Let $\mathcal{V}$ denote the candidate video catalog and $\mathcal{H}_u = (v_{u,1}, \dots, v_{u,L_u})$ the observed interaction history of user u. Our goal is to rank candidate items for the user's next interaction. HGVRec learns a content representation $\mathbf{z}_v \in \mathbb{R}^D$ for each video and a user representation $\mathbf{q}_u \in \mathbb{R}^D$ from the history, with score

$$f(u, v) = \mathbf{q}_u^\top \mathbf{z}_v. \tag{1}$$

The same video encoder supplies history-item and candidate-item representations.

We observe each video through temporally ordered windows of sampled frames. We call an observation window a *scene*; this does not imply a detected shot or a ground-truth action boundary. A frozen MLLM produces a scene graph containing entities, action descriptions, action-specific role assignments, and scene context. After structural validation, video v has S_v valid scenes with original window indices $p_1 < \cdots < p_{S_v}$. Retaining these indices prevents a failed or missing window from silently joining two otherwise separated action segments. An entity has a textual name and attributes; an action has a textual phrase and actor, target, tool, and place assignments. Context is serialized from scene-level format, medium, and topic fields. Entity identifiers specify local graph connectivity and are excluded from text embedding inputs. The video title enters the representation stage as a separate input after visual extraction.

Figure 2 summarizes the computation: REAF constructs role-aware action features, a residual scene encoder produces scene vectors, ASHA organizes them into a video representation, and a sequential recommender models the user's interaction history. The MLLM outputs and their text embeddings remain fixed during recommendation learning. We introduce ASHA first, taking scene vectors as inputs, and then define their construction through REAF. Throughout, $[\cdot\,;\,\cdot]$ denotes concatenation and σ denotes ReLU. Figure 1 illustrates the two levels of structure using a cooking example.

3.2 Action-Segment Hypergraph Aggregation (ASHA)

Action keys and contiguous segments. For a fixed video, let $\mathbf{x}_s \in \mathbb{R}^D$ be the representation of valid scene s and let $\mathcal{A}_s$ be its actions. A deterministic function κ converts an action phrase into a consistency key. The implementation lowercases and tokenizes the phrase, removes participant markers and a fixed set of function words, and normalizes the first remaining token using irregular-form mappings and suffix rules. Empty or unknown phrases produce no key. This is a lightweight lexical rule, not a semantic equivalence test; for example, it does not merge arbitrary synonymous actions. We use the set of keys in a scene,

$$\mathcal{K}_s = \{\kappa(a) \,:\, a \in \mathcal{A}_s,\ \kappa(a) \neq \emptyset\}. \tag{2}$$

Multiple actions with the same key in one scene contribute one membership.

For key k, an action segment e is a maximal run of scenes satisfying both $k \in \mathcal{K}_s$ and consecutiveness of their original indices. Specifically, adjacent members $s, s+1$ must satisfy $p_{s+1} = p_s + 1$. A missing window, a scene without k, or a later reappearance of k separates segments. Singleton segments are retained. A scene containing several keys belongs to several segments, allowing overlapping actions to induce different aggregation groups. Neither participant identity nor temporal identity of entities is required to construct these groups. Thus, a segment expresses continuity of an extracted action key across observations rather than a verified continuous real-world event.

Hypergraph construction. The hypergraph contains one node per valid scene. Its K hyperedges are the action segments, collected in $\mathcal{E}$. Its incidence matrix and associated counts are

$$H_{sk} = \mathbb{I}[s \in e_k], \qquad n_k = \sum_{s=1}^{S} H_{sk}, \qquad d_s = \sum_{k=1}^{K} H_{sk}, \tag{3}$$

where $S = S_v$ and $n_k \geq 1$ for every constructed segment. The incidence structure is fixed before recommendation training. It explicitly records which action key groups which scene indices, providing an inspectable grouping mechanism. This structural property alone does not establish that a group is a faithful explanation of a recommendation.

Scene-to-segment aggregation. Following the two-stage aggregation framework of AllSet [2], we first pool the content of each segment and augment it with its extent:

$$\overline{\mathbf{x}}_k = \frac{1}{n_k} \sum_{s=1}^{S} H_{sk} \mathbf{x}_s, \tag{4}$$

$$\mathbf{l}_k = [\log n_k \,;\, n_k/S], \tag{5}$$

$$\mathbf{u}_k = \sigma\Big(W_{\text{seg}}[\overline{\mathbf{x}}_k \,;\, \mathbf{l}_k] + \mathbf{b}_{\text{seg}}\Big). \tag{6}$$

Here $W_{\text{seg}} \in \mathbb{R}^{D\times(D+2)}$. Mean pooling summarizes segment content, while $\mathbf{l}_k$ preserves information that the mean alone discards. Its entries represent the number of member scenes and its fraction of all valid scenes in the video; neither entry is a duration in seconds.

Segment-to-scene update. A scene receives the mean of the representations of its incident segments:

$$\mathbf{m}_s = \frac{\sum_{k=1}^{K} H_{sk}\mathbf{u}_k}{\max(1, d_s)}, \tag{7}$$

$$\mathbf{x}'_s = \mathbf{x}_s + \alpha\, \mathbb{I}[d_s > 0]\, \sigma\Big(W_{\text{msg}}[\mathbf{x}_s \,;\, \mathbf{m}_s] + \mathbf{b}_{\text{msg}}\Big), \tag{8}$$

where $W_{\text{msg}} \in \mathbb{R}^{D\times 2D}$. Including the receiving scene's own representation allows the update to depend on both local content and segment context. Scenes without valid action keys have $d_s = 0$ and retain $\mathbf{x}_s$. We use one scene-to-segment-to-scene pass.

Video readout. ASHA combines the mean of updated scenes with a separate segment contribution:

$$\overline{\mathbf{u}} = \frac{1}{\max(1, K)} \sum_{k=1}^{K} \mathbf{u}_k, \tag{9}$$

$$\mathbf{z}_v = \frac{1}{S} \sum_{s=1}^{S} \mathbf{x}'_s + \beta\, \mathbb{I}[K > 0] W_{\text{out}} \overline{\mathbf{u}}. \tag{10}$$

$W_{\text{out}} \in \mathbb{R}^{D\times D}$ has no bias. The readout weights segments equally, with segment size already supplied to their encoder in Equation (6). Both α and β are unconstrained trainable scalars initialized to zero; they are shared across videos and are not sigmoid probabilities. At initialization, ASHA therefore reduces to a mean of its input scene vectors. If $K = 0$, its output remains the scene mean. The $S = 0$ case is handled by the title fallback described below.

3.3 Role-aware Entity–Action Fusion (REAF)

Typed features and role assignments. Let $\phi(\cdot) \in \mathbb{R}^F$ be a frozen text encoder. We independently project the video title t_v, entity description e, action phrase a, and scene context c_s into d-dimensional features:

$$\begin{aligned} \mathbf{t}_v &= W_t\phi(t_v) + \mathbf{b}_t, \quad & \mathbf{e}_{s,e} &= W_e\phi(e) + \mathbf{b}_e, \\ \mathbf{a}_{s,a} &= W_a\phi(a) + \mathbf{b}_a, & \mathbf{c}_s &= W_c\phi(c_s) + \mathbf{b}_c. \end{aligned} \tag{11}$$

The entity and action expressions apply to every entity and action in scene s. Missing titles are explicitly masked to zero after projection. Let $\mathcal{R}$ contain four role types: actor, target, tool, and place. For each action a and role $r \in \mathcal{R}$, we obtain a role vector

$$\mathbf{g}_{s,a,r} = \begin{cases} \mathbf{e}_{s,r(a)}, & \text{if role } r \text{ links to entity } r(a), \\ \mathbf{b}_r^{\text{abs}}, & \text{if role } r \text{ is absent}, \\ \mathbf{b}_r^{\text{unk}}, & \text{if role } r \text{ is unknown}. \end{cases} \tag{12}$$

The two state vectors are learned separately for each role. An absent role and an unknown participant therefore need not receive the same representation. REAF uses the extracted assignments; it does not infer new role links or track an entity across scenes.

Action update and scene construction. The residual update of action a in scene s is

$$\begin{aligned} \mathbf{r}_{s,a} &= [\mathbf{g}_{s,a,\text{actor}} \,;\, \mathbf{g}_{s,a,\text{target}} \,;\, \mathbf{g}_{s,a,\text{tool}} \,;\, \mathbf{g}_{s,a,\text{place}}], \\ \tilde{\mathbf{a}}_{s,a} &= \mathbf{a}_{s,a} + \sigma\Big(W_R[\mathbf{a}_{s,a} \,;\, \mathbf{r}_{s,a}] + \mathbf{b}_R\Big), \end{aligned} \tag{13}$$

with $W_R \in \mathbb{R}^{d\times 5d}$. The fixed slot order permits different transformations of the same entity feature when it serves different roles. Entity features themselves are unchanged by this update.

Let $\overline{\mathbf{e}}_s$ be the mean of projected entity features and $\overline{\mathbf{a}}_s$ the mean of updated action features in scene s. The mean of an empty entity or action set is a zero vector. We construct the scene representation as

$$\mathbf{b}_s = [\mathbf{t}_v \,;\, \overline{\mathbf{e}}_s \,;\, \overline{\mathbf{a}}_s \,;\, \mathbf{c}_s], \tag{14}$$

$$\mathbf{x}_s = \mathbf{b}_s + \sigma(W_{\text{scene}}\mathbf{b}_s + \mathbf{b}_{\text{scene}}), \tag{15}$$

where $D = 4d$ and $W_{\text{scene}} \in \mathbb{R}^{D\times D}$. This single-layer residual MLP operates independently on each scene. REAF encodes local participation before pooling actions, while ASHA subsequently models groups of scenes.

Missing observations. Malformed or failed scene outputs do not enter the graph encoder. A valid scene with no actions still contributes its available title, entity, and context information to the scene mean. If a video has no valid scenes, we apply the same residual scene MLP to $[\mathbf{t}_v; \mathbf{0}; \mathbf{0}; \mathbf{0}]$. If neither a title nor a valid scene is available, its final video vector is masked to zero. These rules preserve the recommendation catalog despite extraction failures. They define fallback behavior rather than repair missing visual evidence.

3.4 Sequential Recommendation and Joint Optimization

For each prediction event, we retain the most recent L available history items, add learned position embeddings to their video vectors, and apply a causally masked SASRec-style Transformer [7]. Padding positions are masked. The final normalized hidden state at the last valid position is passed through a two-layer residual user MLP with LayerNorm and GELU to obtain $\mathbf{q}_u$. History positions are modeled chronologically at the user level, independently of ASHA's organization of scenes *inside* an item. Candidate scores follow Equation (1); there is no learned item-ID embedding table in the content encoder.

Popularity-corrected in-batch objective. A training batch contains B history–target pairs $(\mathcal{H}_i, y_i)$. The targets of other batch examples serve as negatives. Let $Q(v)$ be the normalized frequency of v as a target among training transitions with nonempty histories; validation and test targets are excluded from this frequency estimate. The corrected logit for user-history row i and target column j is

$$\ell_{ij} = \mathbf{q}_i^\top \mathbf{z}_{y_j} - \log Q(y_j). \tag{16}$$

For $j \neq i$, we mask column j if $y_j = y_i$ or y_j occurs in the truncated input history $\mathcal{H}_i$. The diagonal positive is always retained. Writing $\tilde{\ell}_{ij}$ for the resulting logits, the objective is

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\tilde{\ell}_{ii})}{\sum_{j=1}^{B} \exp(\tilde{\ell}_{ij})}. \tag{17}$$

The implementation sets masked logits to -10^4. The popularity correction is used during training; evaluation ranks the uncorrected dot-product scores. All typed projections, role-state embeddings, REAF, the scene MLP, ASHA, the sequential encoder, and the user MLP are optimized jointly. The frozen MLLM and text encoder receive no recommendation gradients, and action-segment memberships are not learned.

4 Experiments

We use results from run `R05_v8_261105`. HGVRec is evaluated on two datasets through three questions:

- **RQ1: Overall effectiveness.** How does HGVRec compare with title-only, scene-description, and mean-pooled scene-graph representations?
- **RQ2: Adding action segments.** What changes when ASHA is added to the scene-graph encoder?
- **RQ3: Adding role information.** What changes when REAF is added to the encoder that already uses ASHA?

These questions evaluate the recommendation utility of the available extracted information and the incremental contribution of the two modules.

4.1 Experimental Setup

Datasets. We use MicroLens-100K for its micro-video interactions and video content [9]. We also use M^{3}L-10M, which adds multimodal movie information, including trailers, to MovieLens-10M [11]. Our M^{3}L setup uses titles and available trailer videos; every rating is treated as an interaction without weighting by the rating value. Movies with unavailable or unusable videos and their interactions are excluded during source preparation. Accordingly, its prediction task concerns the next movie in a rating-event sequence. Rating timestamps do not establish the order in which the user watched the movies or their trailers.

For both datasets, users must have at least five source interactions. We retain at most their 13 most recent events, ordered by timestamp with source-event order breaking ties. The candidate catalog is the union of items in these retained sequences. The last event of each user is held out for test and the penultimate event for validation. Earlier events supply training transitions, excluding the first event, which has no preceding history. Table 1 reports the resulting evaluation cohorts. The counts were reconstructed from the available preparation events and accepted only after their split hashes matched the stored R05 split for the corresponding dataset.

Table 1: Retained evaluation cohorts. Training transitions exclude each user's first event and the two held-out events. Validation and test contain one event per user.

Statistic	MicroLens-100K	M^{3}L-10M
Users	100,000	69,878
Candidate items	19,671	8,197
Retained interactions	678,355	908,412
Training transitions	378,355	698,778
Validation events	100,000	69,878
Test events	100,000	69,878

Prediction and ranking protocol. Training examples use only the history preceding their target event. Validation selects the checkpoint with the highest HR@10, retaining the first checkpoint in a tie. Test uses the selected checkpoint and the history preceding the test event, including the validation interaction; there is no refit on validation labels. The model accepts a truncated history, while evaluation masks *all* items in the preceding retained history before ranking the full catalog. The masking rule also applies if the held-out target repeats an earlier item. Scores are sorted in descending order using the recorded evaluation contract. There is no sampled-negative evaluation. This user-wise split does not impose a shared global time boundary across users.

Representation variants. Table 2 maps the five archived configurations to their paper names. All use a content-based sequential recommendation backbone. Title-only projects the title embedding and applies a residual item MLP. Scene Description combines a projected title with projected scene-description embeddings, applies a residual scene MLP, and averages the scenes. Scene Graph projects title, entity, action, and context features; it pools entities and actions separately within each scene and averages the encoded scenes. Scene Graph + ASHA adds the action-segment module to this graph encoder. HGVRec further adds REAF before scene pooling. Parameters are trained independently for each variant and seed.

The three graph variants have identical stored graph-input and representation hashes within each dataset. Their comparison therefore reuses the same extracted graph information. The description and graph sources have their own extracted outputs and valid-scene sets, so their difference also includes the extraction format and input encoding. Videos without valid extracted scenes use the fallback described in Section 3.3 rather than being dropped from a variant's candidate catalog.

Implementation and configuration provenance. The archived source labels identify Qwen3-VL graph and description extraction and a BGE-based text-feature representation. The shared graph inputs are held fixed across the graph comparisons, and the learned representation modules are optimized with the recommender. The inspected implementation uses AdamW with weight decay 0.1 and gradient-norm clipping at 5.0 for the objective in Section 3.4.

Table 2: Compared content encoders and archived active trainable parameter counts, including the sequential recommender. Frozen extraction and text-embedding models are excluded. Counts are identical across the two datasets.

Method	Content input	REAF	ASHA	Trainable parameters
Title-only	Title	–	–	7,624,704
Scene Description	Title + scene descriptions	–	–	7,624,704
Scene Graph	Title + scene graphs	–	–	7,624,704
Scene Graph + ASHA	Title + scene graphs	–	Yes	8,675,330
HGVRec	Title + scene graphs	Yes	Yes	8,758,402

These implementation details describe the available code; the historical run configuration still requires completion as noted below. The stored experiments use seeds 42, 43, and 44, giving 15 trained variant–seed combinations per dataset.

Draft reproducibility note. The available R05 archive contains the diagnostic results and parameter reports but not the original training and extraction configuration files. The exact historical MLLM and text-encoder checkpoint revisions, prompt versions, frame/window settings, embedding and history dimensions, learning rate, batch size, epoch limit, patience, precision, and hardware remain to be verified. We therefore leave those run-specific settings unresolved rather than infer them from the repository's current defaults.

Metrics and statistical analysis. We report the mean and sample standard deviation of HR@10 and NDCG@10 over the three seeds. For a single held-out target of rank r, the per-user metrics are

$$\begin{aligned} \mathrm{HR@10}(r) &= \mathbb{I}[r \le 10], \\ \mathrm{NDCG@10}(r) &= \frac{\mathbb{I}[r \le 10]}{\log_2(r+1)}. \end{aligned} \tag{18}$$

Relative improvement over reference method b is $100(M_a/M_b - 1)$ for metric M. For NDCG@10, the archived analysis first averages each user's metric over seeds and then performs paired bootstrap resampling over users with equal user weight and 10,000 replicates. Each dataset has a separate family of all ten pairwise comparisons among its five variants. Bonferroni correction at family-wise level 0.05 yields a 99.5% confidence interval for each difference. Table 4 displays four relevant contrasts while preserving this original ten-comparison correction. We use the stored intervals without rerunning the bootstrap. The reported standard deviations describe seed variation; they are distinct from the user-bootstrap confidence intervals. HR@10 is reported descriptively, without significance claims.

4.2 Overall Recommendation Performance (RQ1)

Table 3 shows that HGVRec achieves the highest mean NDCG@10 and HR@10 among the five compared variants on both datasets. On MicroLens-100K, it reaches NDCG@10 of 0.043448 and HR@10 of 0.082210, compared with 0.033708 and 0.064190 for Title-only. These correspond to relative improvements of 28.90% and 28.07%, respectively. On M^3L-10M, HGVRec reaches 0.143475 and 0.240467, improving over Title-only by 3.38% and 3.45%. The HGVRec–Title-only NDCG@10 intervals exclude zero on both datasets (Table 4).

Scene Description improves NDCG@10 over Title-only by 12.56% on MicroLens-100K and 0.68% on M^3L-10M; only the former difference is significant under the stored correction. HGVRec also improves over Scene Description, by 14.51% and 2.68% in NDCG@10, respectively, with positive corrected intervals for both comparisons. Thus, the complete structured encoder provides a measurable advantage over the description-based variant under this evaluation protocol.

Scene Graph alone is below Scene Description on both datasets: its NDCG@10 deficits are 0.002799 and 0.001597, respectively, and both are significant in the archived pairwise analysis. On M^3L-10M, neither Scene Graph nor Scene Description differs significantly from Title-only in NDCG@10. These observations indicate that producing graph-formatted outputs and pooling their components is insufficient by itself to reproduce the gains of HGVRec. Because descriptions and graphs are different extracted inputs, this comparison does not isolate a single encoding operation; the graph-to-graph comparisons below more directly assess the added modules.

4.3 Effect of Action-Segment Aggregation (RQ2)

Adding ASHA to Scene Graph raises NDCG@10 from 0.035142 to 0.042218 on MicroLens-100K and from 0.138132 to 0.142260 on M^3L-10M. The absolute increases are 0.007076 and 0.004128, corresponding to relative gains of 20.14% and 2.99%. Both corrected confidence intervals are positive. Mean HR@10 also increases, from 0.068667 to 0.080500 and from 0.232362 to 0.239799, respectively. Since the graph variants share their extracted inputs, this comparison supports the usefulness of adding action-segment-based aggregation to the otherwise mean-pooled scene encoder.

The intervention adds the whole ASHA module: action-segment membership, size-aware segment encoding, scene updates, and segment readout. It also adds 1,050,626 trainable parameters. The present comparison therefore measures the contribution of that complete module rather than independently identifying the effects of hyperedge structure, size features, or the two message/readout paths. The reference encoder averages scenes and does not encode their chronology. Consequently, this experiment does not establish that action segments outperform a chronological scene encoder or an attention-based scene aggregator.

Table 3: Recommendation performance, reported as mean ± sample standard deviation across seeds 42, 43, and 44. Relative changes use Title-only within each dataset. Bold indicates the highest mean, not an additional significance test.

Method	NDCG@10	HR@10	ΔNDCG (%)	ΔHR (%)
MicroLens-100K				
Title-only	0.033708 ± 0.000474	0.064190 ± 0.000925	–	–
Scene Description	0.037941 ± 0.001164	0.072610 ± 0.001585	+12.56	+13.12
Scene Graph	0.035142 ± 0.001069	0.068667 ± 0.001931	+4.25	+6.97
Scene Graph + ASHA	0.042218 ± 0.001034	0.080500 ± 0.001876	+25.25	+25.41
HGVRec	**0.043448** ± 0.000429	**0.082210** ± 0.001040	+28.90	+28.07
M^{3}L-10M				
Title-only	0.138784 ± 0.001909	0.232443 ± 0.002425	–	–
Scene Description	0.139729 ± 0.001739	0.234041 ± 0.001755	+0.68	+0.69
Scene Graph	0.138132 ± 0.000708	0.232362 ± 0.000673	-0.47	-0.03
Scene Graph + ASHA	0.142260 ± 0.001667	0.239799 ± 0.001916	+2.50	+3.16
HGVRec	**0.143475** ± 0.000611	**0.240467** ± 0.000638	+3.38	+3.45

Table 4: Selected NDCG@10 contrasts from the archived paired-user bootstrap. Differences are left minus right; relative changes use the right-hand method. All intervals retain Bonferroni correction over the original ten comparisons per dataset.

Contrast	Absolute difference	Relative (%)	99.5% confidence interval
MicroLens-100K			
HGVRec − Title-only	+0.009740	+28.90	[0.008874, 0.010591]
HGVRec − Scene Description	+0.005507	+14.51	[0.004708, 0.006299]
Scene Graph + ASHA − Scene Graph	+0.007076	+20.14	[0.006338, 0.007803]
HGVRec − Scene Graph + ASHA	+0.001230	+2.91	[0.000534, 0.001916]
M^{3}L-10M			
HGVRec − Title-only	+0.004691	+3.38	[0.003377, 0.005948]
HGVRec − Scene Description	+0.003746	+2.68	[0.002458, 0.005013]
Scene Graph + ASHA − Scene Graph	+0.004128	+2.99	[0.002910, 0.005336]
HGVRec − Scene Graph + ASHA	+0.001215	+0.85	[0.000026, 0.002420]

4.4 Effect of Role-aware Fusion (RQ3)

Adding REAF to Scene Graph + ASHA increases NDCG@10 by 0.001230 on MicroLens-100K and 0.001215 on M^{3}L-10M, corresponding to relative gains of 2.91% and 0.85%. The corrected intervals are $[0.000534, 0.001916]$ and $[0.000026, 0.002420]$, respectively. Both exclude zero, although the lower bound for M^{3}L-10M is close to zero. Mean HR@10 increases from 0.080500 to 0.082210 and from 0.239799 to 0.240467. REAF adds 83,072 trainable parameters to the ASHA-equipped model.

This result is consistent with encoding actor, target, tool, and place assignments before pooling actions being useful for the subsequent action-segment representation. The contrast holds the extracted graphs and presence of ASHA fixed, while changing the role-fusion computation and its parameters. R05 does not include a REAF-only variant; it therefore supports the incremental benefit of REAF *in the presence of ASHA*, but does not measure its standalone contribution or a factorial interaction between the modules.

4.5 Interpretation and Evaluation Scope

The relative HGVRec gain over Title-only is larger on MicroLens-100K, but both the baseline level and the absolute improvement differ. Title-only NDCG@10 is 0.033708 on MicroLens-100K and 0.138784 on M^{3}L-10M, while the corresponding absolute HGVRec gains are 0.009740 and 0.004691. The retained cohorts also differ in catalog size and history length (Table 1), as well as in content and interaction semantics. These descriptive differences do not identify the cause of the cross-dataset performance gap. The stored graph scene-coverage rates are similar, 96.04% and 95.92%, but extraction coverage is not a measure of semantic correctness or role grounding.

The evidence establishes improvements among the five evaluated variants using the archived extraction sources. The three graph configurations provide a controlled comparison of how the same graph information is used by the learned encoder, subject to their differing parameter counts. The results do not directly measure correction of hallucinations, fidelity of recommendation explanations, or performance on an on-device accelerator. Our fixed structural memberships are inspectable, while their explanatory value requires a separate evaluation.

The broader questions raised in the Introduction remain open where R05 lacks the necessary comparisons: chronological scene encoding, scene attention, stronger MLLM extractors, and device latency or memory consumption. No conclusion about scaling with MLLM capability or a measured on-device efficiency gain is drawn from this run. Completing those comparisons and the

run-specific configuration record is necessary before expanding the empirical claims beyond the reported results.

5 Conclusion

We presented HGVRec, a video representation learning framework that organizes MLLM-extracted semantics through explicit structure. ASHA groups scenes into maximal contiguous action segments and aggregates their information through a hypergraph, while REAF encodes entity participation roles within each scene. Experiments on MicroLens-100K and M^3L-10M show that HGVRec achieves the highest mean NDCG@10 and HR@10 among the five evaluated variants. With shared graph inputs, adding ASHA improves NDCG@10 on both datasets, and adding REAF provides further gains when ASHA is present. These findings support explicit role and action-segment modeling as a way to improve the recommendation utility of available MLLM outputs. Future work will examine comparisons with chronological scene encoders and attention, stronger MLLM extractors, and device-level efficiency to assess the framework's broader applicability.

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